1. Perform data quality checks by checking for missing values, if any.

In [115…
```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import numpy as np

from sklearn.cluster import KMeans
from sklearn.model_selection import train_test_split

from imblearn.over_sampling import SMOTE
from sklearn.linear_model import LogisticRegression
from sklearn.ensemble import RandomForestClassifier,GradientBoostingClassifi

from sklearn.model_selection import cross_val_score
from sklearn.metrics import classification_report, confusion_matrix
from sklearn.metrics import r2_score, auc, roc_curve, roc_auc_score

import warnings
warnings.filterwarnings('ignore')
```

In [116…
```
emp_data = pd.read_csv('data/HR_comma_sep.csv')
```

In [117…
```
emp_data.head()
```

Out[117…

	satisfaction_level	last_evaluation	number_project	average_montly_hours	time_sp
0	0.38	0.53	2	157	
1	0.80	0.86	5	262	
2	0.11	0.88	7	272	
3	0.72	0.87	5	223	
4	0.37	0.52	2	159	

In [118…
```
emp_data.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 14999 entries, 0 to 14998
Data columns (total 10 columns):
 #   Column                 Non-Null Count  Dtype
---  ------                 --------------  -----
 0   satisfaction_level     14999 non-null  float64
 1   last_evaluation        14999 non-null  float64
 2   number_project         14999 non-null  int64
 3   average_montly_hours   14999 non-null  int64
 4   time_spend_company     14999 non-null  int64
 5   Work_accident          14999 non-null  int64
 6   left                   14999 non-null  int64
 7   promotion_last_5years  14999 non-null  int64
 8   sales                  14999 non-null  str
 9   salary                 14999 non-null  str
dtypes: float64(2), int64(6), str(2)
memory usage: 1.3 MB
```

In [119…
```
emp_data.describe()
```

Out[119…

	satisfaction_level	last_evaluation	number_project	average_montly_hours	time
count	14999.000000	14999.000000	14999.000000	14999.000000	
mean	0.612834	0.716102	3.803054	201.050337	
std	0.248631	0.171169	1.232592	49.943099	
min	0.090000	0.360000	2.000000	96.000000	
25%	0.440000	0.560000	3.000000	156.000000	
50%	0.640000	0.720000	4.000000	200.000000	
75%	0.820000	0.870000	5.000000	245.000000	
max	1.000000	1.000000	7.000000	310.000000	

In [120…
```
emp_data.shape
```

Out[120… `(14999, 10)`

In [121…
```
emp_data.isna().sum()
```

Out[121…
```
satisfaction_level       0
last_evaluation          0
number_project           0
average_montly_hours     0
time_spend_company       0
Work_accident            0
left                     0
promotion_last_5years    0
sales                    0
salary                   0
dtype: int64
```

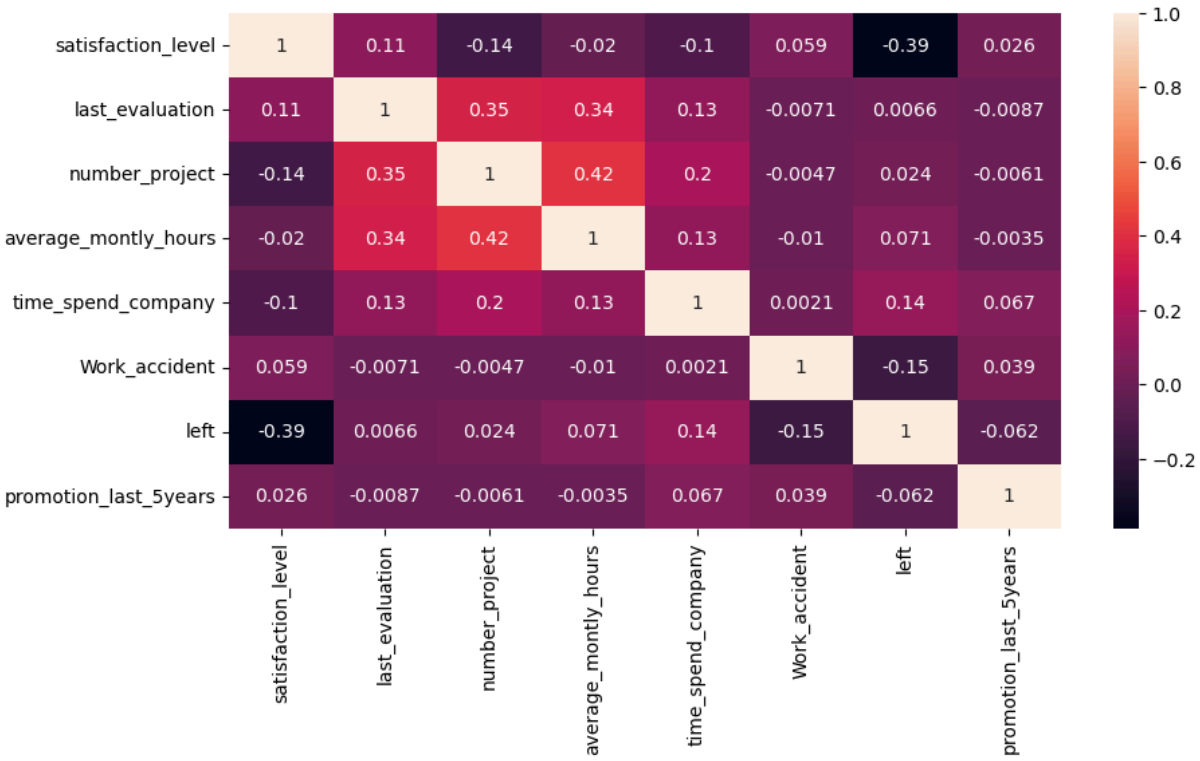
Observation

Not having any missing values in the given data set

2. Understand what factors contributed most to employee turnover at EDA.

2.1 Draw a heatmap of the correlation matrix between all numerical features or columns in the data.

In [122…
```
plt.figure(figsize=(10,5))
sns.heatmap(emp_data.iloc[:,0:8].corr(), annot=True)
plt.show()
```

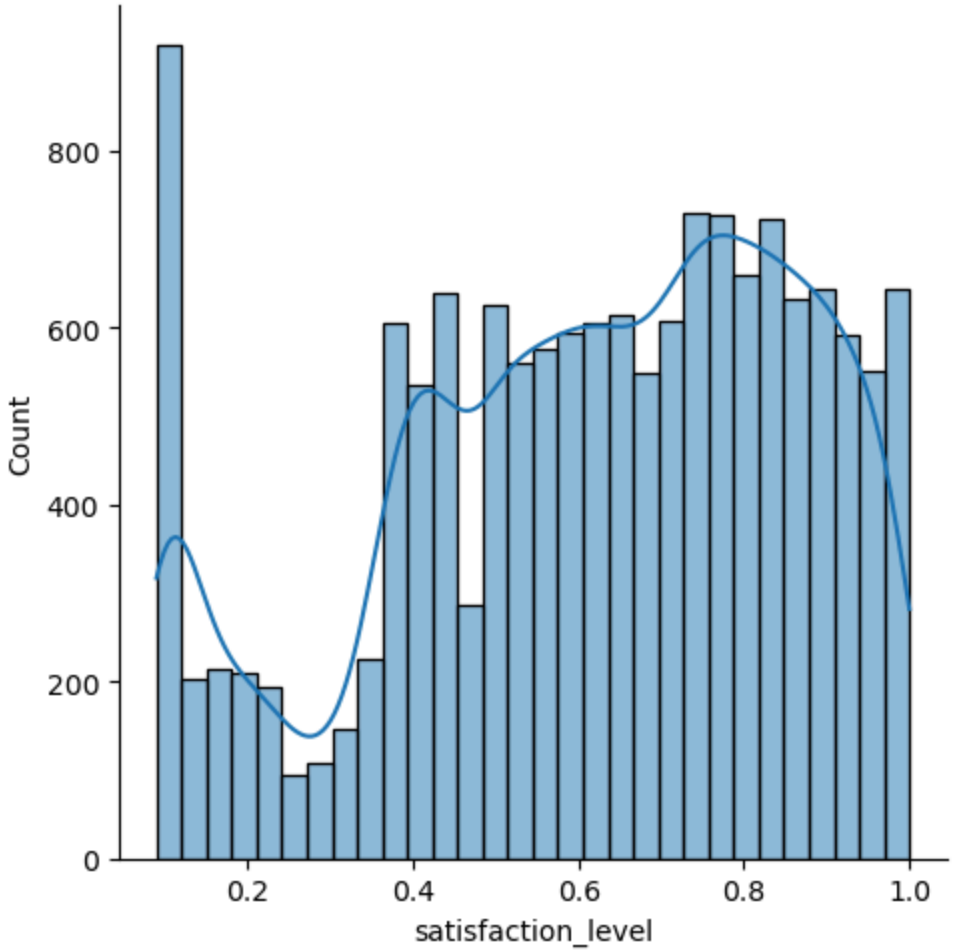


2.2 Draw the distribution plot of:

Employee Satisfaction (use column satisfaction_level)

In [123…
```
plt.figure(figsize=(10,5))
sns.displot(emp_data['satisfaction_level'], kde=True)
plt.show()
```

```
<Figure size 1000x500 with 0 Axes>
```

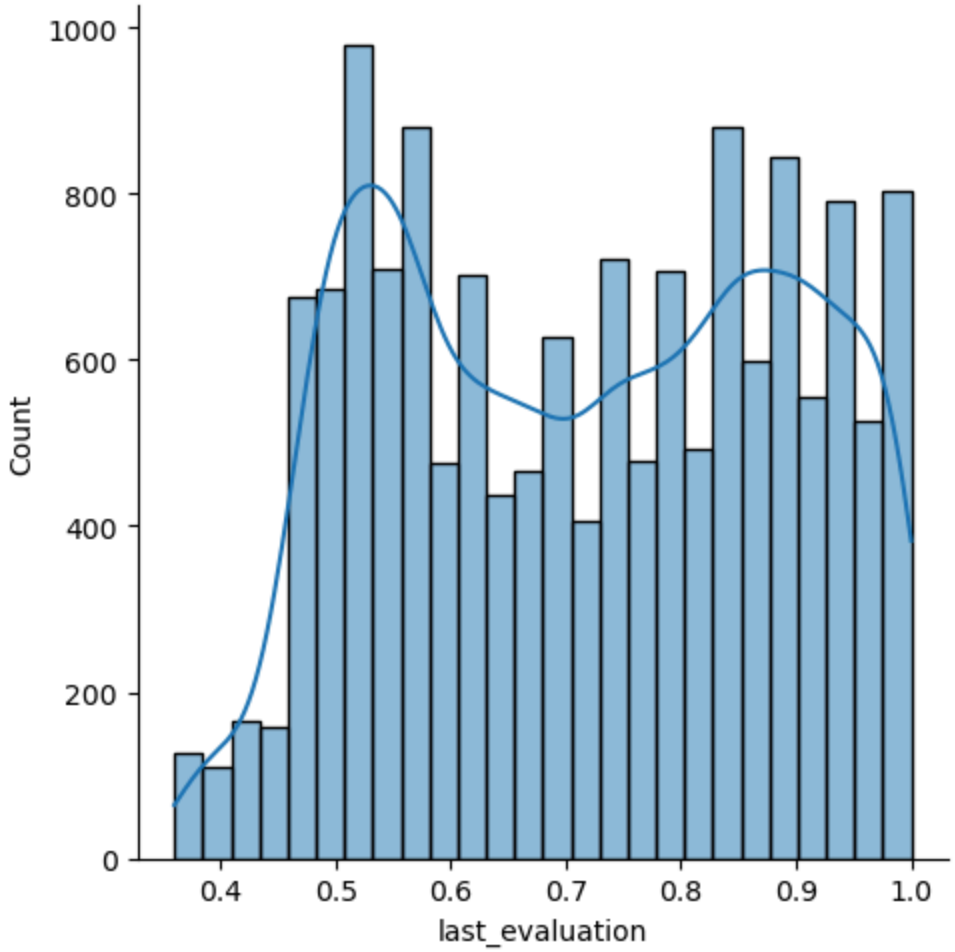


Employee Evaluation (use column last_evaluation)

In [124…

```
plt.figure(figsize=(10,5))
sns.displot(emp_data['last_evaluation'], kde=True)
plt.show()
```

```
<Figure size 1000x500 with 0 Axes>
```

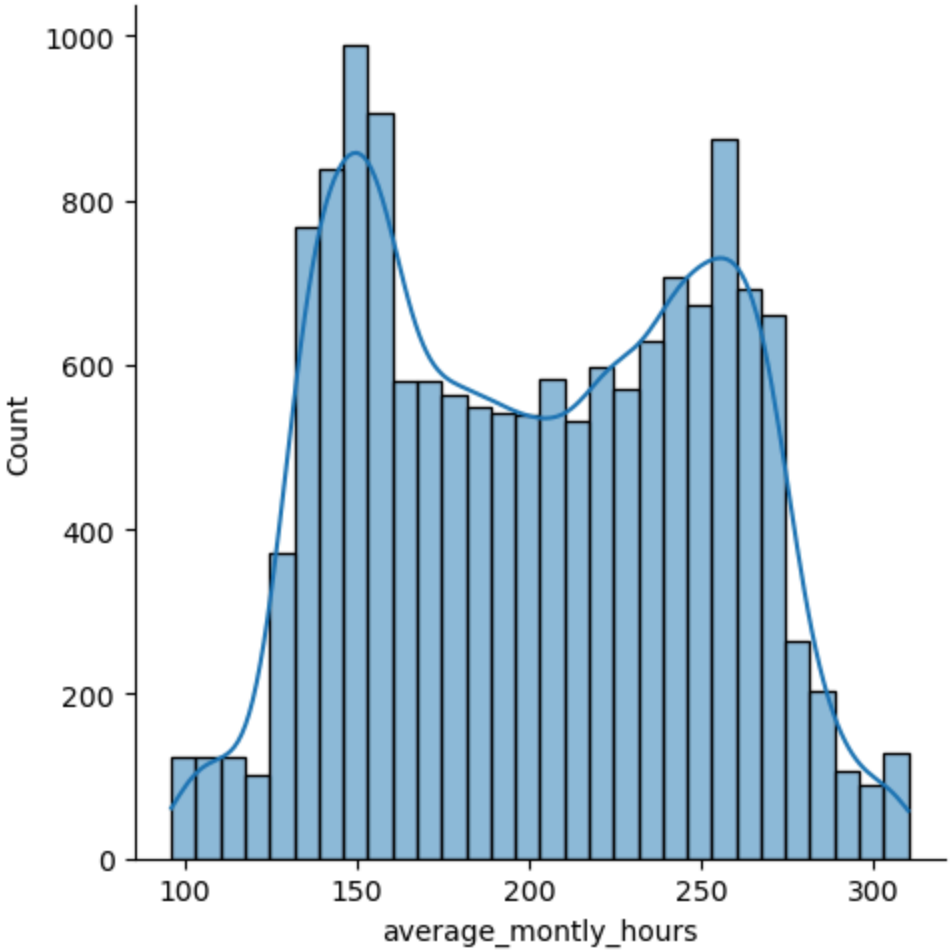


Employee Average Monthly Hours (use column average_montly_hours)

In [125…

```
plt.figure(figsize=(10,5))
sns.displot(emp_data['average_montly_hours'], kde=True)
plt.show()
```

```
<Figure size 1000x500 with 0 Axes>
```



2.3 Draw the bar plot of the employee project count of both employees who left and stayed in the organization (use column number_project and hue column left), and give your inferences from the plot.

In [126…
```
emp_data.head()
```

Out[126…

	satisfaction_level	last_evaluation	number_project	average_montly_hours	time_sp
0	0.38	0.53	2	157	
1	0.80	0.86	5	262	
2	0.11	0.88	7	272	
3	0.72	0.87	5	223	
4	0.37	0.52	2	159	

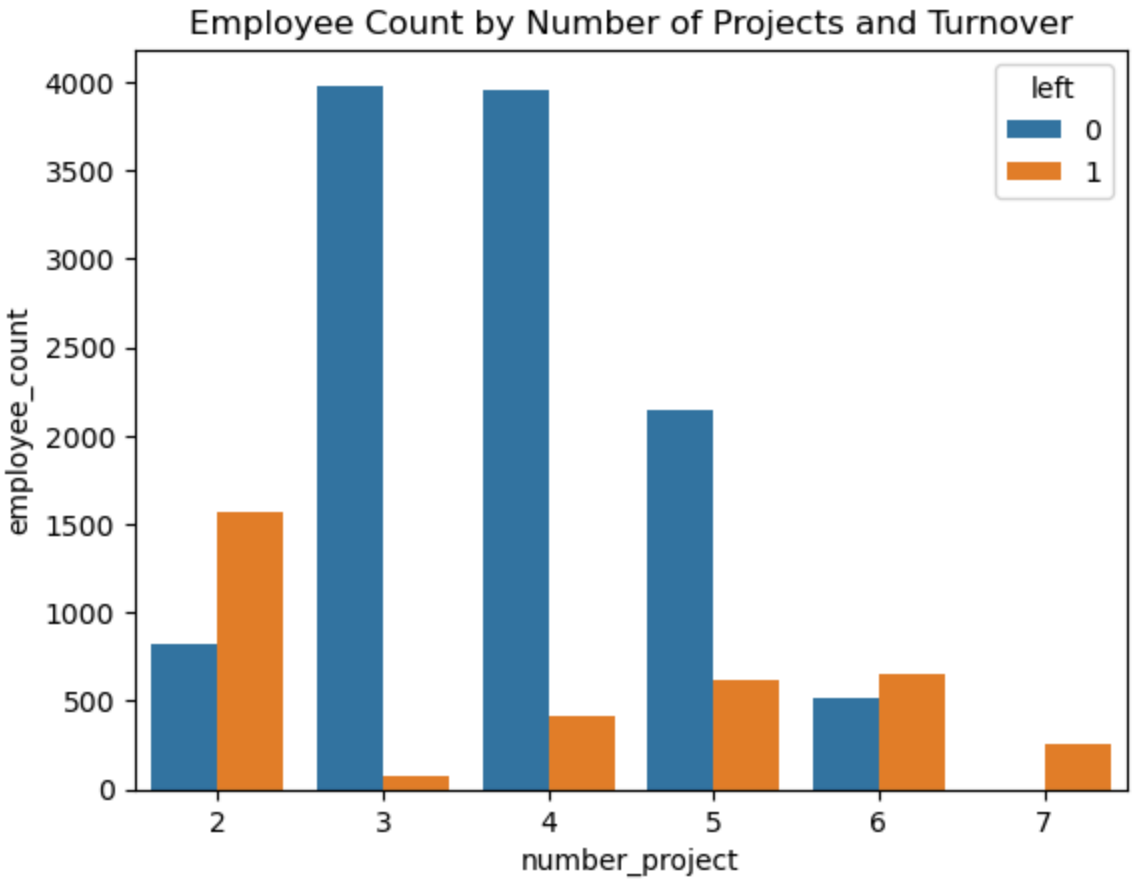
In [127…
```
emp_project_left = emp_data.groupby(['number_project','left'])['number_proje
emp_project_left
```

Out [127...

	number_project	left	employee_count
0	2	0	821
1	2	1	1567
2	3	0	3983
3	3	1	72
4	4	0	3956
5	4	1	409
6	5	0	2149
7	5	1	612
8	6	0	519
9	6	1	655
10	7	1	256

In [128...

```
sns.barplot(x='number_project', y='employee_count',hue='left', data=emp_proj
plt.title('Employee Count by Number of Projects and Turnover')
plt.show()
```



Observations:

- Employees who is having 5 to 7 project count and heavy work load and high turnover rate.
- Employees who is having 3 - 4 project count is medium work load and low turnover rate.
- Employees who is having 2 project count who is having less work load and high turnover rate.

Optimal project count to low turnover rate is 3 to 4 projects. Suggested for good work life balance.

3. Perform clustering of employees who left based on their satisfaction and evaluation.

3.1 Choose columns satisfaction_level, last_evaluation, and left.

In [129…
```
emp_data_left = emp_data[emp_data['left']==1][['satisfaction_level','last_ev
emp_data_left
```

Out[129…

	satisfaction_level	last_evaluation
0	0.38	0.53
1	0.80	0.86
2	0.11	0.88
3	0.72	0.87
4	0.37	0.52
...	...	...
14994	0.40	0.57
14995	0.37	0.48
14996	0.37	0.53
14997	0.11	0.96
14998	0.37	0.52

3571 rows × 2 columns

3.2 Do K-means clustering of employees who left the company into 3 clusters?

In [130…
```
kmeans = KMeans(n_clusters=3,random_state=42)
```

In [131…
```
kmeans.fit(emp_data_left)
```

Out[131…

▾ KMeans ⓘ ?

▸ Parameters

In [132…
```
kmeans.inertia_
```

Out[132…
```
27.007707973602802
```

In [133…
```
kmeans.cluster_centers_
```

Out[133…
```
array([[0.41014545, 0.51698182],
       [0.80851586, 0.91170931],
       [0.11115466, 0.86930085]])
```

In [134…
```
kmeans.labels_
```

Out[134…
```
array([0, 1, 2, ..., 0, 2, 0], shape=(3571,), dtype=int32)
```

In [135…
```
emp_data_left['cluster'] = kmeans.labels_
```

In [136…
```
emp_data_left
```

Out[136…

	satisfaction_level	last_evaluation	cluster
0	0.38	0.53	0
1	0.80	0.86	1
2	0.11	0.88	2
3	0.72	0.87	1
4	0.37	0.52	0
...	...	...	...
14994	0.40	0.57	0
14995	0.37	0.48	0
14996	0.37	0.53	0
14997	0.11	0.96	2
14998	0.37	0.52	0

3571 rows × 3 columns

3.4 Based on the satisfaction and evaluation factors, give your thoughts on the employee clusters.

In [137…
```
centers = pd.DataFrame(kmeans.cluster_centers_, columns=['satisfaction_level
centers
```

Out [137…

	satisfaction_level	last_evaluation
0	0.410145	0.516982
1	0.808516	0.911709
2	0.111155	0.869301

In [138…

```
sns.scatterplot(data = emp_data_left, x = 'satisfaction_level' , y = 'last_e
sns.scatterplot(x = centers['satisfaction_level'], y = centers['last_evaluat
plt.title('Kmeans Clustering of left employees')
plt.show()
```

Observation

Cluster 0:

Moderate Satisfaction, Moderate Evaluation
These employees are generally dissatisfied from their jobs.
They have a least evaluation.
They left the company might be because of dissatisfaction or poor ratings & feedback from manager.

Based on above points, HR can take some action to maintain positive environment.

```
Cluster 1:

    High Satisfaction, High Evaluation
    They are fully satisfied with their roles and tasks
    They are seems top performers.
    They might left due to some personal causes or might be
they are holding offers from other company

Cluster 2:

    Low Satisfaction, High Evaluation
    They are seems top performers.
    These employees are also one of the top performers but
are unsatisfied with their current roles and assignments.
    They left might be due they are not getting promotions,
lack of recognition in the company or no career growth option

    Based on above points, HR and project team may start
upskilling employees by conducting training programs. Can
initiate Rewards and Recognition programs.
```

4. Handle the left Class Imbalance using the SMOTE technique.

4.1 Pre-process the data by converting categorical columns to numerical columns by:

Separating categorical variables and numeric variables

Applying get_dummies() to the categorical variables

Combining categorical variables and numeric variables

In [139…
```
emp_data = pd.get_dummies(emp_data, columns=['salary','sales'], drop_first =
emp_data
```

Out[139…

	satisfaction_level	last_evaluation	number_project	average_montly_hours	tim
0	0.38	0.53	2	157	
1	0.80	0.86	5	262	
2	0.11	0.88	7	272	
3	0.72	0.87	5	223	
4	0.37	0.52	2	159	
...	...	...	...	...	
14994	0.40	0.57	2	151	
14995	0.37	0.48	2	160	
14996	0.37	0.53	2	143	
14997	0.11	0.96	6	280	
14998	0.37	0.52	2	158	

14999 rows × 19 columns

In [140…

```
emp_data.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 14999 entries, 0 to 14998
Data columns (total 19 columns):
 #   Column                 Non-Null Count  Dtype
---  ------                 --------------  -----
 0   satisfaction_level     14999 non-null  float64
 1   last_evaluation        14999 non-null  float64
 2   number_project         14999 non-null  int64
 3   average_montly_hours   14999 non-null  int64
 4   time_spend_company     14999 non-null  int64
 5   Work_accident          14999 non-null  int64
 6   left                   14999 non-null  int64
 7   promotion_last_5years  14999 non-null  int64
 8   salary_low             14999 non-null  bool
 9   salary_medium          14999 non-null  bool
 10  sales_RandD            14999 non-null  bool
 11  sales_accounting       14999 non-null  bool
 12  sales_hr               14999 non-null  bool
 13  sales_management       14999 non-null  bool
 14  sales_marketing        14999 non-null  bool
 15  sales_product_mng      14999 non-null  bool
 16  sales_sales            14999 non-null  bool
 17  sales_support          14999 non-null  bool
 18  sales_technical        14999 non-null  bool
dtypes: bool(11), float64(2), int64(6)
memory usage: 1.1 MB
```

4.2 Do the stratified split of the dataset to train and test in the ratio 80:20 with random_state=123.

In [141…
```
x = emp_data.drop(['left'], axis=1)
y = emp_data['left']
```

In [142…
```
x_train,x_test,y_train,y_test = train_test_split(x,y,test_size=20,random_sta
```

4.3 Upsample the train dataset using the SMOTE technique from the imblearn module.

In [143…
```
emp_data['left'].value_counts()
```

Out[143…
```
left
0    11428
1     3571
Name: count, dtype: int64
```

In [144…
```
smt = SMOTE(sampling_strategy='minority')
x_train_rs, y_train_rs = smt.fit_resample(x_train, y_train)
```

In [145…
```
y_train_rs.value_counts()
```

Out[145…
```
left
0    11412
1    11412
Name: count, dtype: int64
```

5. Perform 5-fold cross-validation model training and evaluate performance

5.1 Train a logistic regression model, apply a 5-fold CV, and plot the classification report.

In [146…
```
logistic_model = LogisticRegression()
logistic_model.fit(x_train_rs,y_train_rs)
y_pred_lom = logistic_model.predict(x_test)
```

In [147…
```
cv_score_lom = cross_val_score(LogisticRegression(), x_train_rs, y_train_rs,
print(cv_score_lom)
print(np.mean(cv_score_lom))
```

```
[0.74107338 0.76122673 0.81007667 0.78751369 0.7971078 ]
0.7793996542300186
```

In [148…
```
cr_lom = classification_report(y_test, y_pred_lom)
print("Classification Report of Logistic Model : ")
print(cr_lom)
```

```
Classification Report of Logistic Model :
              precision    recall  f1-score   support

           0       0.89      1.00      0.94        16
           1       1.00      0.50      0.67         4

    accuracy                           0.90        20
   macro avg       0.94      0.75      0.80        20
weighted avg       0.91      0.90      0.89        20
```

5.2 Train a Random Forest Classifier model, apply the 5-fold CV, and plot the classification report.

In [149…
```
random_forest = RandomForestClassifier()
random_forest.fit(x_train_rs,y_train_rs)
y_pred_rfc = random_forest.predict(x_test)
```

In [150…
```
cv_score_rfc = cross_val_score(RandomForestClassifier(), x_train_rs, y_trair
print(cv_score_rfc)
print(np.mean(cv_score_rfc))
```

```
[0.98313253 0.98598028 0.9864184  0.98598028 0.98575811]
0.9854539214942794
```

In [151…
```
cr_rfc = classification_report(y_test, y_pred_rfc)
print("Classification Report of Random Forest Classifier : ")
print(cr_rfc)
```

```
Classification Report of Random Forest Classifier :
              precision    recall  f1-score   support

           0       1.00      1.00      1.00        16
           1       1.00      1.00      1.00         4

    accuracy                           1.00        20
   macro avg       1.00      1.00      1.00        20
weighted avg       1.00      1.00      1.00        20
```

5.3 Train a Gradient Boosting Classifier model, apply the 5-fold CV, and plot the classification report.

In [152…
```
gradient_boosting = GradientBoostingClassifier()
gradient_boosting.fit(x_train_rs,y_train_rs)
y_pred_gbc = gradient_boosting.predict(x_test)
```

In [153…
```
cv_score_gbc = cross_val_score(GradientBoostingClassifier(), x_train_rs, y_t
print(cv_score_gbc)
print(np.mean(cv_score_gbc))
```

```
[0.95596933 0.9618839  0.96254107 0.9686747  0.96209465]
0.9622327314196631
```

In [154…
```
cr_gbc = classification_report(y_test, y_pred_gbc)
print("Classification Report of Gradient Boosting Classifier : ")
```

```
print(cr_gbc)
```

```
Classification Report of Gradient Boosting Classifier :
              precision    recall  f1-score   support

           0       1.00      1.00      1.00        16
           1       1.00      1.00      1.00         4

    accuracy                           1.00        20
   macro avg       1.00      1.00      1.00        20
weighted avg       1.00      1.00      1.00        20
```

6. Identify the best model and justify the evaluation metrics used.

6.1 Find the ROC/AUC for each model and plot the ROC curve.

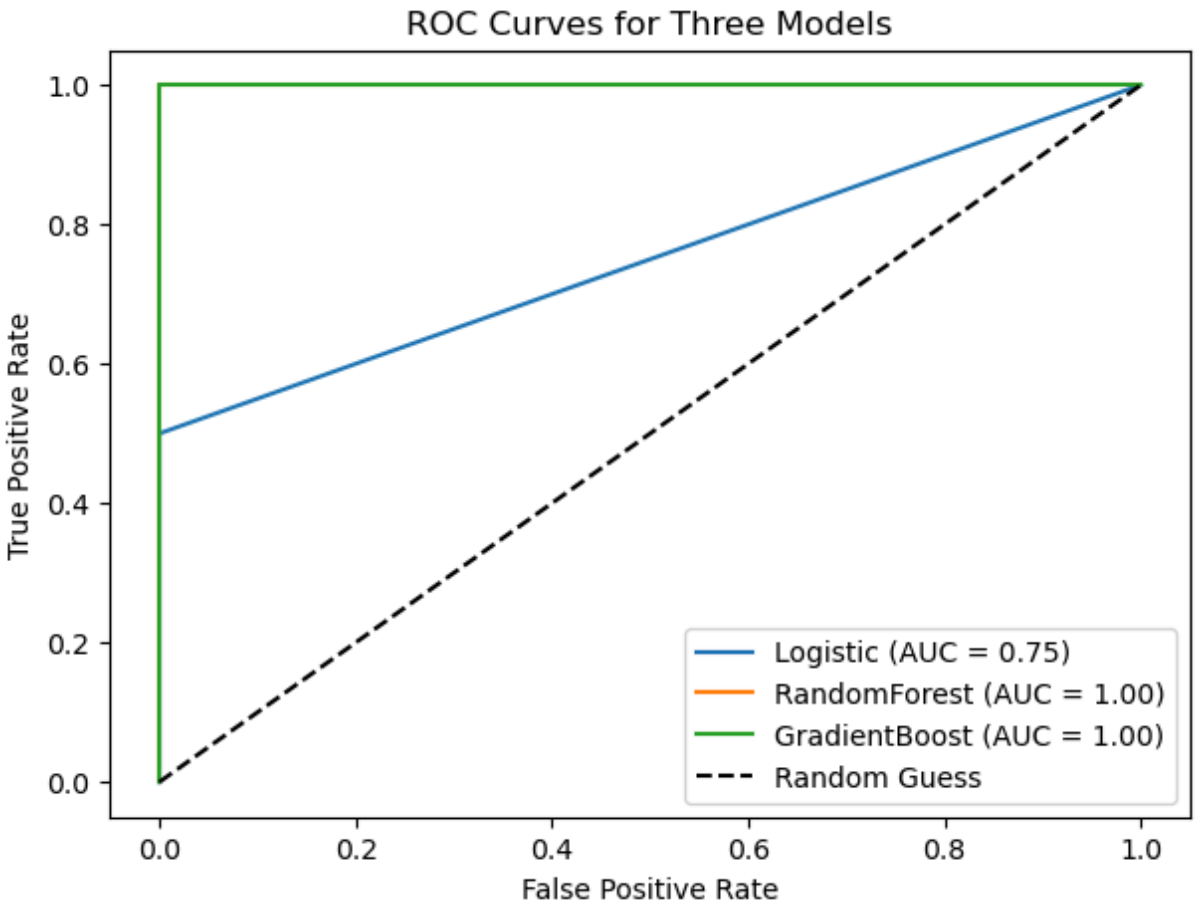
In [155…
```
models_df = pd.DataFrame({'Test': y_test, 'Logistic': y_pred_lom, 'RandomFor
models_df
```

Out [155…

	Test	Logistic	RandomForest	GradientBoost
6958	0	0	0	0
7534	0	0	0	0
2975	0	0	0	0
3903	0	0	0	0
8437	0	0	0	0
6812	0	0	0	0
1567	1	1	1	1
14679	1	0	1	1
10188	0	0	0	0
11718	0	0	0	0
7040	0	0	0	0
4130	0	0	0	0
6872	0	0	0	0
10889	0	0	0	0
4659	0	0	0	0
1896	1	1	1	1
4915	0	0	0	0
603	1	0	1	1
7680	0	0	0	0
9865	0	0	0	0

In [156…

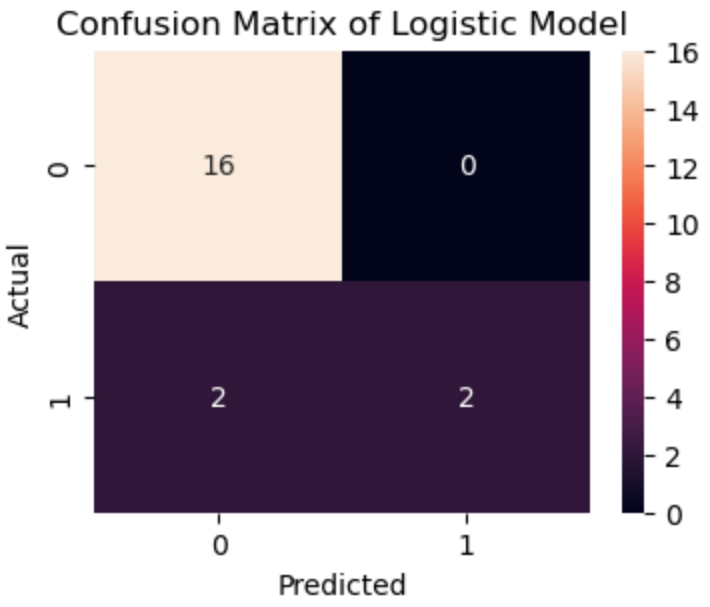
```
# Plotting ROC curve
plt.figure(figsize=(7, 5))

for model in ['Logistic', 'RandomForest', 'GradientBoost']:
    fpr, tpr, _ = roc_curve(models_df['Test'], models_df[model])
    auc = roc_auc_score(models_df['Test'], models_df[model])
    plt.plot(fpr, tpr, label=f'{model} (AUC = {auc:.2f})')

plt.plot([0, 1], [0, 1], 'k--', label='Random Guess')

plt.xlabel('False Positive Rate')
plt.ylabel('True Positive Rate')
plt.title('ROC Curves for Three Models')
plt.legend()
plt.show()
```

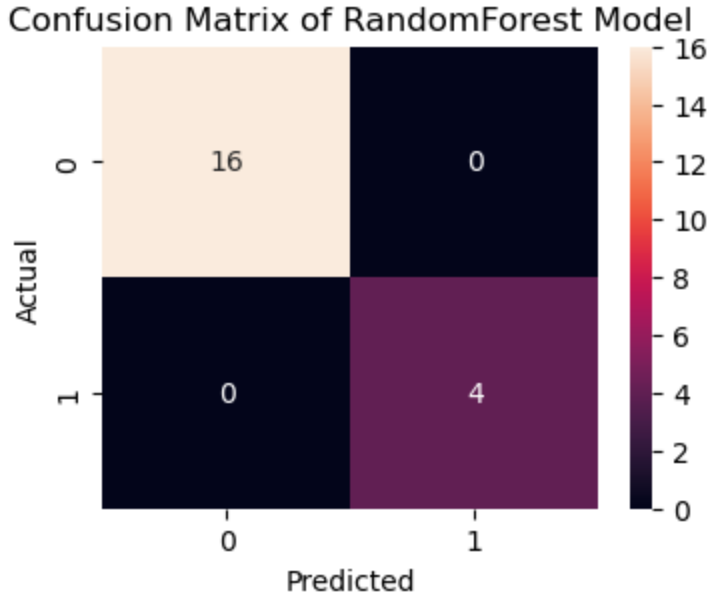


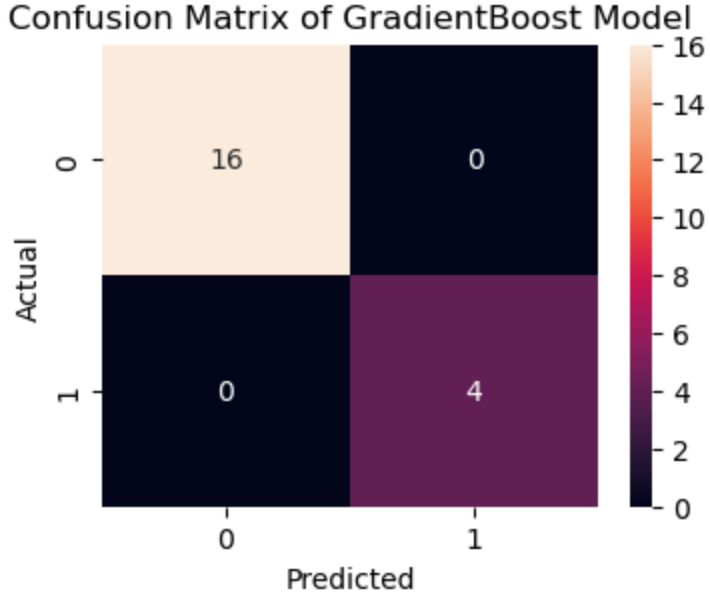
6.2 Find the confusion matrix for each of the models.

In [159…

```
for model in ['Logistic', 'RandomForest', 'GradientBoost']:
  cm = confusion_matrix(models_df['Test'], models_df[model])
  plt.figure(figsize=(4,3))
  sns.heatmap(cm, annot=True, fmt='d')
  plt.title('Confusion Matrix of ' + model + ' Model')
  plt.xlabel('Predicted')
  plt.ylabel('Actual')
  plt.show()
```


Confusion Matrix of Logistic Model
16
0
2
2
Actual
Predicted
0
1
0
1
16
14
12
10
8
6
4
2
0


Confusion Matrix of RandomForest Model
16
0
0
4
Actual
Predicted
0
1
0
1
16
14
12
10
8
6
4
2
0


Confusion Matrix of GradientBoost Model
16
0
0
4
Actual
Predicted
0
1
0
1
16
14
12
10
8
6
4
2
0

6.3 Explain which metric needs to be used from the confusion matrix: Recall or Precision?

Recall is more useful. Failing to identify employees who are leaving company is more critical. So Recall should be useful over precision. High recall ensures that the majority of employees who are likely to leave are identified.

7. Suggest various retention strategies for targeted employees.

7.1 Using the best model, predict the probability of employee turnover in the test data.

In [166…

```
# from Randomforest classifier model already got the predicted y
rfc_test_data = x_test.copy()
rfc_test_data['prob_of_leaving'] = y_pred_rfc
rfc_test_data
```

Out [166…

	satisfaction_level	last_evaluation	number_project	average_montly_hours	tim
6958	0.54	0.67	3	154	
7534	0.72	0.52	3	143	
2975	0.95	0.61	3	267	
3903	0.78	0.79	3	203	
8437	0.60	0.40	3	146	
6812	0.78	0.65	3	157	
1567	0.40	0.48	2	132	
14679	0.90	0.92	5	245	
10188	0.87	0.49	4	213	
11718	0.63	0.85	4	182	
7040	0.86	0.65	4	134	
4130	0.81	0.58	3	243	
6872	0.66	0.86	4	112	
10889	0.76	0.57	3	148	
4659	0.82	0.97	5	235	
1896	0.39	0.49	2	127	
4915	0.65	0.78	4	238	
603	0.83	0.90	5	245	
7680	0.72	0.89	4	217	
9865	0.96	0.51	5	205	

7.2 Based on the probability score range below, categorize the employees into four zones and suggest your thoughts on the retention strategies for each zone.

Safe Zone (Green) (Score < 20%)

Low-Risk Zone (Yellow) (20% < Score < 60%)

Medium-Risk Zone (Orange) (60% < Score < 90%)

High-Risk Zone (Red) (Score > 90%).

In [167…

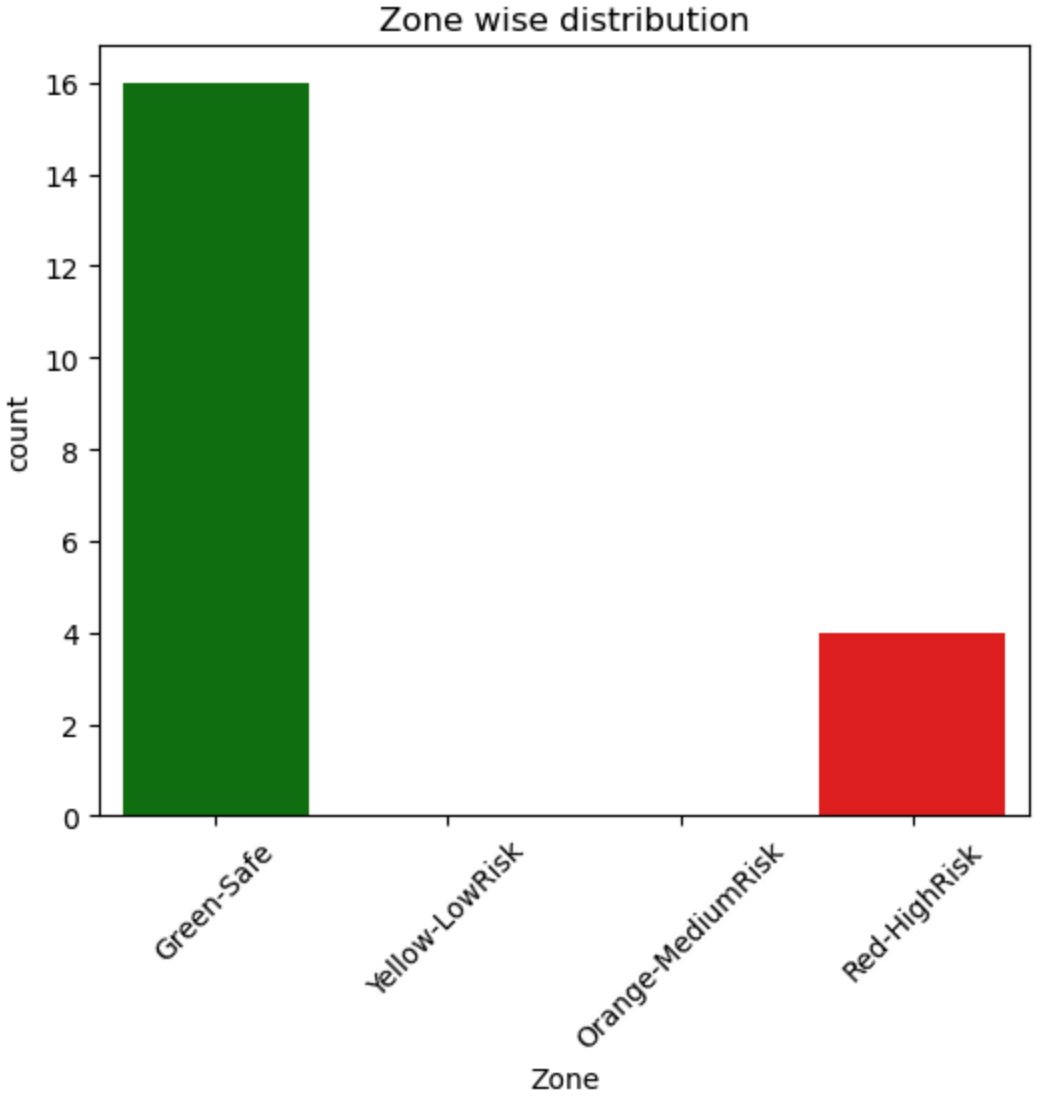
```
rfc_test_data['Zone'] = pd.cut(rfc_test_data['prob_of_leaving'], bins=[0.0,
=True)
rfc_test_data
```

Out[167...

	satisfaction_level	last_evaluation	number_project	average_montly_hours	tim
6958	0.54	0.67	3	154	
7534	0.72	0.52	3	143	
2975	0.95	0.61	3	267	
3903	0.78	0.79	3	203	
8437	0.60	0.40	3	146	
6812	0.78	0.65	3	157	
1567	0.40	0.48	2	132	
14679	0.90	0.92	5	245	
10188	0.87	0.49	4	213	
11718	0.63	0.85	4	182	
7040	0.86	0.65	4	134	
4130	0.81	0.58	3	243	
6872	0.66	0.86	4	112	
10889	0.76	0.57	3	148	
4659	0.82	0.97	5	235	
1896	0.39	0.49	2	127	
4915	0.65	0.78	4	238	
603	0.83	0.90	5	245	
7680	0.72	0.89	4	217	
9865	0.96	0.51	5	205	

In [168…

```
plt.figure(figsize=(6,5))
sns.countplot(x='Zone', data=rfc_test_data, palette=['Green', 'Yellow', 'Ora
plt.title('Zone wise distribution')
plt.xticks(rotation=45)
plt.show()
```



Observations

HR can continue existing policies, as employees are in their most comfortable zone. Low-Risk Zone: HR can work on starting quarterly rewards or recognition programmes to motivate employees. Medium-Risk Zone: HR can conduct training programmes to upskill employees into their area of interest. and can assign challenging roles as per their experience. High-Risk Zone: HR can offer target-based designation promotion and can have one-on-one talks with employees to discuss their expectations.

In [ ]: